Computational Text Analysis

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Course Description and Rationale	With vast amounts of administrative, legislative, political or news texts available, computational text analysis (also known as text-as-data) has gained increasing traction in social science research over the past years. This two-day workshop introduces PhD researchers to computational text analysis methods from a public policy and public administration research perspective. Whether you study policies, politics, economies, organisations, or institutions, texts are critical traces of societal changes and conflicts. This workshop equips you with practical skills to extract meaningful insights from such textual data at scale, moving beyond what manual analysis alone can achieve. The workshop emphasizes hands-on application. Through interactive exercises using R and the quanteda package, you'll learn to implement methods ranging from dictionary-based approaches to machine learning techniques. By the end, you'll be prepared to incorporate computational text analysis into your own research. The seminar is organized in three stages of the learning process. First, participants will learn about the *foundations* of text-as-data in the social sciences. This includes the role of texts in political science and public administration and the theoretical underpinnings of computational text analysis. Next, we will focus on the *methods* of computational text analysis. Participants will learn about the use of text-as-data methods through hands-on exercises using the statistical programming language R. We will also discuss recent examples of empirical research using computational text analysis. Last but not least, participants will *apply* the learned methods to their own research or interests. Participants can either bring their own textual data or use data provided by the instructor and subsequently implement one or several of the techniques learned throughout the seminar.

Objectives	The course is designed as an introduction to computational text analysis. Participants will: **Theoretical Foundations**: I. Learn about the theoretical underpinnings of text-as-data approaches and their relevance in social science research **Practical Application**: II. Develop practical skills in implementing text-as-data methods using the R statistical programming language through hands-on exercises **Transfer to Own Research**: III. Apply the learned methods to your own area of research either by developing a first analysis on data in your field or by critically assessing research using text-as-data in your field
Reading and seminar preparation	Before the first session, participants are asked to read one article and one chapter: • Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). Text as Data: A New Framework for Machine Learning and the Social Sciences. Chapter 1. • Martin K (2018): Imagine All the People: Literature, Society, and Cross-National Variation in Education Systems. World Politics 70(3). https://doi.org/10.1017/S0043887118000023 There is also another small assignment before class which is to install R, RStudio and the package 'quanteda'. You can find a quick guide on how to install R and RStudio here: https://rstudio-education.github.io/hopr/starting.html. If you do not manage, reach out to milan.thies@unil.ch. Further optional reading: • General introductory reading: Benoit, K. (2020). Text as data: An overview. In L. Curini & R. Franzese (Eds.), The SAGE Handbook of Research Methods in Political Science and International Relations. SAGE Publications Ltd. https://doi.org/10.4135/9781526486387

	• On LDA Topic Models: Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *The journal of machine learning research*, 993–1022 • On Seeded LDA: Watanabe, K., & Zhou, Y. (2022). Theory-Driven Analysis of Large Corpora: Semisupervised Topic Classification of the UN Speeches. *Social science computer review*, 40(2), 346–366. https://doi.org/10.1177/0894439320907027 • On Natural Language Inference: Laurer, M., Van Atteveldt, W., Casas, A., & Welbers, K. (2024). Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and bert-nli. *Political Analysis*, *32*(1), 84–100. • On preprocessing: Denny, M., & Spirling, A. (2017). Text Preprocessing for Unsupervised Learning: Why It Matters, When It Misleads, and What to Do about It. https://doi.org/10.2139/ssrn.2849145 Examples of recent applications: • Kittel, R. C. (2025). Simply Speaking? Language Complexity among (Non-)Populist Actors in Parliamentary Debates. *Government and Opposition*, *60*(4), 1336–1366. https://doi.org/10.1017/gov.2025.4 • Hurka, S., Knill, C., & Steinebach, Y. (2024). Rules as policy data? Measuring and linking policy substance and legislative context. *Regulation & Governance*, *18*(3), 724–739. https://doi.org/10.1111/rego.12528 • Thies, M. (2025). State Intervention in Vocational Education: Training for the Digital and Green Transitions. *Regulation & Governance*. https://doi.org/10.1111/rego.70074 • Carstensen, M. B., Ibsen, C. L., Jensen, I. M. N., & Lund-Sørensen, B. (2025). Mapping Green Skills in Collective Skill Formation Systems: A Natural Language Processing Analysis of Danish Vocational Education and Training. *Regulation & Governance*. https://doi.org/10.1111/rego.70097

Sessions	
Foundations of Computational Text Analysis	
Day 1 **Session 1**	**Texts as Traces of Political Conflict and Policy Change** • Overview of the course; Expectations • Textual manifestation of political conflict • Textual manifestations of policy change • Reflection: What kind of data is text in my own research interests?
Day 1 **Session 2**	**A Newcomer's Guide to Computational Text Analysis** • Fundamental assumptions of computational text analysis • Varieties of text-as-data techniques • Strengths and weaknesses of computational text analysis
Methods of Computational Text Analysis	
Day 1 **Session 3**	**Getting Started: Introduction to R for Text Analysis, Quanteda & Descriptive Analyses** • R for text analysis • Representation of text in Quanteda • Importing textual data and preprocessing • Descriptive analyses (word frequencies, keywords in context, keyness, word clouds)
Day 1 **Session 4**	**Computational Text Analysis Without Machine Learning** • Word counts in social science research • Dictionaries as a text-as-data technique

Day 2 **Session 1**	**Supervised- and Unsupervised Machine Learning for Text Analysis** • Overview: unsupervised and supervised machine learning • Unsupervised topic models • (Semi-)supervised topic models
Application	
Day 2 **Session 2**	**Developing a Text-as-Data Research Project** • Recent applications • How to develop a text-as-data research project • Researchers are supported in developing their own computational text analysis project. Focus at this stage: Selection of data and methods fit for research question
Day 2 **Session 3**	**Supervised Project Work** • Students apply several techniques learned throughout the seminar • During the project students will be assisted in overcoming difficulties by the instructor
Day 2 **Session 4**	**Presentation, Reflection and Outlook** • Presentation: Most interesting finding of each in-class project. • Reflection: Learning from the seminar & how computational text analysis techniques could be used in own research • Outlook: Recent advances in text-as-data research